

ICQ LABS

Investment Club Quant Labs

Content Requirements Document

Investment Club at the University of Central Florida

Version 1.0

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1. Project Overview

ICQ Labs (Investment Club Quant Labs) is the internal web platform for Investment Club at the University of Central Florida. It serves as the centralized hub for portfolio management, financial research, stock analysis, collaboration, and education for all club members.

Platform Goals

- Provide real-time visibility into the club's portfolio holdings and performance
- Deliver a curated news feed directly correlated to current holdings
- Offer interactive financial models (DCF, regression, comps) for stock analysis
- Facilitate structured stock pitch submissions and member voting
- Enable role-based collaboration between members, officers, and administrators
- Serve as an educational resource for quantitative finance concepts

Target Users

The platform will serve three distinct user groups: general club members (view and interact), officers (submit research and manage content), and the admin (full platform control). The expected user base is the active membership of Investment Club UCF.

2. User Roles & Permissions

ICQ Labs enforces a three-tier role-based access control system. Each role inherits the permissions of the roles below it, with the admin role having unrestricted access to all platform features.

2.1 Role Definitions

Member: General club members. Can view all public content (portfolio, news, research reports, models). Can interact with content through voting on pitches, adding stocks to personal watchlists, and participating in discussions. Cannot submit research, modify models, or manage platform settings.

Officer: Club officers and analysts. Inherits all Member permissions. Can submit research reports and stock pitch decks. Can create and edit meeting notes. Can modify parameters on research models. Cannot manage members, execute trades, or access admin settings.

Admin: Platform administrator (single user). Unrestricted access to all features. Can manage member accounts and role assignments. Can add/remove portfolio holdings, execute trades, configure platform settings, and access the audit log.

2.2 Permission Matrix

Feature / Action	Section	Member	Officer	Admin	Notes
View portfolio holdings	Portfolio	✓	✓	✓	
View performance metrics	Portfolio	✓	✓	✓	
Add/remove holdings	Portfolio	—	—	✓	Admin only
Execute trades	Portfolio	—	—	✓	Admin only
Read news feed	News	✓	✓	✓	
Filter/search news	News	✓	✓	✓	
View research reports	Research	✓	✓	✓	
Submit research reports	Research	—	✓	✓	
Edit/delete any research	Research	—	—	✓	
Run DCF model	Models	✓	✓	✓	View only
Run regression analysis	Models	✓	✓	✓	View only
Modify model parameters	Models	—	✓	✓	
Submit stock pitches	Pitches	—	✓	✓	
Vote on stock pitches	Pitches	✓	✓	✓	
View meeting notes	Meetings	✓	✓	✓	
Create meeting notes	Meetings	—	✓	✓	
View member directory	Members	✓	✓	✓	
Manage members/roles	Members	—	—	✓	Admin only
View audit log	Settings	—	—	✓	Admin only
Configure platform	Settings	—	—	✓	Admin only

3. Authentication & Access

The platform requires authentication for all users. The login system must support the three role tiers and enforce permissions across the entire application.

Element	Description	Data Source / Notes
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Login page	Branded login screen with ICQ Labs logo, email/password fields, and role indicator after login	Custom auth or OAuth
Member login	Standard authentication for general club members; grants view and interact permissions	Email + password
Officer login	Authenticated login that grants elevated permissions for content submission and model editing	Email + password + role flag
Admin login	Single admin account with full platform access; should have additional security (2FA recommended)	Email + password + 2FA
Session management	Persistent sessions with configurable timeout; role displayed in the navigation bar at all times	JWT or session cookies
Password reset	Self-service password reset flow via email	Email verification
Registration	Admin-controlled; new members are added by the admin and receive an invite email with temporary credentials	Admin panel action

4. Portfolio Dashboard

The portfolio dashboard is the primary landing page after login. It provides a comprehensive view of the club's current holdings, allocation, and performance metrics. All members can view; only the admin can modify holdings.

4.1 Holdings Table

Element	Description	Data Source / Notes
Holdings table	Sortable table showing all current positions: ticker, company name, shares held, average cost, current price, P/L (\$), P/L (%), and portfolio weight	Market data API
Sector breakdown	Pie or donut chart showing portfolio allocation by sector (Technology, Financials, Healthcare, etc.)	Derived from holdings
Total portfolio value	Prominent display of aggregate portfolio value, total cost basis, total unrealized P/L, and daily change	Calculated real-time
Individual position cards	Expandable cards per holding with mini price chart (1D, 1W, 1M, 3M, 1Y), key stats (P/E, market cap, 52-week range), and link to full research	Market data API

4.2 Performance Tracking

Element	Description	Data Source / Notes
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Historical performance chart	Interactive line chart showing portfolio value over time with adjustable date range (1W, 1M, 3M, 6M, YTD, 1Y, ALL)	Calculated from trade history
Benchmark comparison	Overlay performance against S&P 500, QQQ, and Russell 2000 on the same chart	Index data API
Performance metrics	Display Sharpe ratio, annualized return, max drawdown, beta vs S&P 500, and alpha	Calculated
Trade history log	Complete log of all buy/sell transactions with date, ticker, action, shares, price, and who executed	Internal database

4.3 Admin Controls (Admin Only)

Element	Description	Data Source / Notes
Add position	Form to add a new holding: ticker, shares, purchase price, purchase date	Admin action
Close position	Sell/remove a holding with sale price and date; moves to trade history	Admin action
Adjust position	Modify share count or average cost for an existing holding	Admin action
Cash balance	Track and display available cash in the portfolio	Manual entry

5. News Feed

A curated, real-time news feed that is directly correlated to the stocks currently held in the portfolio. The feed should only surface news relevant to the club’s positions, with the ability to filter and search.

Element	Description	Data Source / Notes
Holdings-correlated feed	Automated news aggregation filtered to only show articles mentioning current portfolio tickers or their parent companies	News API (e.g., Finnhub, Alpha Vantage, NewsAPI)
Sentiment indicator	Each article tagged with a sentiment score or label (Bullish / Neutral / Bearish) based on NLP analysis	Sentiment analysis model
Source attribution	Display source name (Reuters, Bloomberg, CNBC, etc.), publication time, and direct link to the original article	News API metadata
Ticker filter	Dropdown or tag-based filter to view news for a specific holding or multiple holdings	Frontend filter
Search	Full-text search across all news articles in the feed	Search index

Earnings calendar	Integrated calendar or banner showing upcoming earnings dates for all holdings with countdown timers	Earnings data API
SEC filing alerts	Notifications when new 10-K, 10-Q, or 8-K filings are published for any holding	SEC EDGAR API
Macro news section	Separate tab or section for macroeconomic news (Fed decisions, CPI releases, employment data) that affects the overall portfolio	News API + economic calendar

6. Research Models

Interactive financial models that club members can use to analyze current and prospective holdings. Members can view and run models with default parameters; officers and admin can modify model inputs and save configurations.

6.1 Discounted Cash Flow (DCF) Model

Element	Description	Data Source / Notes
Revenue projections	Input fields for projected revenue growth rates over a 5–10 year forecast period	Officer/Admin editable
Margin assumptions	Configurable gross margin, operating margin, and net margin assumptions per year	Officer/Admin editable
WACC calculator	Built-in weighted average cost of capital calculator with inputs for risk-free rate, equity risk premium, beta, cost of debt, and capital structure	Auto-populated with defaults
Terminal value	Toggle between perpetuity growth method and exit multiple method with adjustable parameters	Officer/Admin editable
Sensitivity table	Two-variable sensitivity analysis (e.g., WACC vs. terminal growth rate) with color-coded output grid	Auto-generated
Fair value output	Clear display of implied share price vs. current market price with upside/downside percentage	Calculated
Save/load scenarios	Officers and admin can save named scenarios for each stock; members can view saved scenarios	Internal database

6.2 Regression Analysis

Element	Description	Data Source / Notes
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Single regression	Simple linear regression of a stock's returns against a chosen independent variable (market return, sector ETF, etc.)	Market data API
Multi-factor regression	Multiple regression with selectable factors: market return, SMB, HML, momentum, and custom factors	Factor data source
Output display	Regression coefficients, R-squared, adjusted R-squared, p-values, residual plots, and confidence intervals	Calculated
Time period selector	Adjustable date range for the regression analysis (1Y, 3Y, 5Y, custom)	Market data API
Visualization	Scatter plot with regression line, residual histogram, and Q-Q plot	Charting library

6.3 Comparable Company Analysis (Comps)

Element	Description	Data Source / Notes
Peer group selection	Search and select comparable companies by ticker, sector, or market cap range	Market data API
Comps table	Side-by-side table of key multiples: P/E, EV/EBITDA, P/S, P/B, EV/Revenue, PEG ratio, and dividend yield	Financial data API
Median/mean benchmarks	Calculated median and mean for each multiple across the peer group, with the target highlighted	Calculated
Implied valuation	Implied share price for the target stock based on each multiple and the peer group median	Calculated

6.4 Technical Analysis

Element	Description	Data Source / Notes
Interactive price chart	Candlestick or line chart with adjustable timeframes (1D, 1W, 1M, 3M, 6M, 1Y, 5Y)	Market data API
Technical indicators	Overlay common indicators: SMA (20, 50, 200), EMA, RSI, MACD, Bollinger Bands, VWAP, and volume bars	Calculated from price data
Drawing tools	Trendlines, horizontal support/resistance levels, and Fibonacci retracement tools	Frontend implementation
Indicator settings	Adjustable parameters for each indicator (e.g., SMA period, RSI lookback)	Frontend controls

6.5 Options Pricing (Optional/Future)

Element	Description	Data Source / Notes
Black-Scholes calculator	Input fields for stock price, strike, time to expiry, volatility, and risk-free rate; outputs call/put fair value and Greeks	Calculated
Options chain viewer	Real-time options chain with bid/ask, volume, open interest, and implied volatility for each strike	Options data API
Payoff diagrams	Visual payoff diagram for single-leg and multi-leg strategies	Calculated + charting

7. Stock Pitch System

A structured workflow for officers to submit stock pitch proposals, for members to review and vote, and for admin to approve or reject additions to the portfolio.

Element	Description	Data Source / Notes
Pitch submission form	Structured form: ticker, investment thesis (rich text), target price, time horizon, risk factors, catalysts, and supporting model link	Officer/Admin only
Pitch deck upload	File upload for PDF or PowerPoint pitch decks that display inline	File storage
Voting system	Members and officers can vote Buy / Hold / Pass on each pitch with optional comments; results displayed as aggregate bar chart	Internal database
Pitch status workflow	Status labels: Submitted → Under Review → Voted → Approved / Rejected; admin controls status transitions	Admin controlled
Pitch archive	Historical archive of all past pitches with outcome tracking (was the stock added? how did it perform after?)	Internal database
Discussion thread	Comment thread on each pitch for members to ask questions and discuss the thesis	Internal database
Deadline tracking	Voting deadline per pitch with countdown timer; automatic status change when deadline passes	Automated

8. Watchlist

A personal and shared watchlist feature for tracking stocks that the club is considering but has not yet added to the portfolio.

Element	Description	Data Source / Notes
Personal watchlist	Each member can maintain a private watchlist of tickers with current price, daily change, and mini sparkline chart	Market data API
Club watchlist	Shared watchlist visible to all members; officers and admin can add/remove stocks	Internal database
Price alerts	Set custom price alerts (above/below threshold) for any watchlist stock; notification via email or in-app	Market data API + notifications
Quick stats	Watchlist view shows P/E, market cap, 52-week range, and average volume for each ticker	Market data API
Add to pitch	One-click action to start a pitch submission for a watchlist stock (officer/admin only)	Internal workflow

9. Risk & Analytics

Advanced portfolio analytics and risk management tools that give the club a quantitative edge in understanding their exposure and performance attribution.

Element	Description	Data Source / Notes
Correlation matrix	Heatmap showing pairwise correlation between all holdings over a selectable time period	Calculated from returns
Value at Risk (VaR)	Portfolio-level VaR at 95% and 99% confidence intervals using historical simulation and parametric methods	Calculated
Beta exposure	Portfolio beta vs. S&P 500 and breakdown of beta contribution per holding	Calculated
Sector concentration	Visual display of sector weights with warning indicators if any sector exceeds a configurable threshold	Derived from holdings
Efficient frontier	Mean-variance optimization chart showing the current portfolio vs. the efficient frontier with optimal allocation suggestions	Calculated
Drawdown analysis	Chart showing peak-to-trough drawdowns over time with max drawdown highlighted	Calculated from portfolio NAV
Performance attribution	Breakdown of portfolio return into allocation effect, selection effect, and interaction effect vs. benchmark	Calculated

10. Meetings & Collaboration

Tools for managing club meetings, documenting decisions, and facilitating collaboration between members.

Element	Description	Data Source / Notes
Meeting notes	Rich text editor for officers to create and publish meeting minutes with date, attendees, agenda, and action items	Officer/Admin only to create
Meeting archive	Chronological archive of all past meeting notes, searchable by date or keyword	Internal database
Action item tracker	Extracted action items from meeting notes with assignee, deadline, and completion status	Internal database
Trade journal	Shared journal where officers and admin document the rationale for each trade: thesis, entry/exit criteria, and post-trade review	Officer/Admin to create
Announcements	Pinned announcements banner on the dashboard for important club-wide updates (officer/admin only to create)	Internal database

11. Member Management

Admin-only tools for managing the club's member roster, role assignments, and platform access.

Element	Description	Data Source / Notes
Member directory	Searchable list of all members with name, email, role, join date, and active/inactive status (visible to all)	Internal database
Role management	Admin can assign or change roles (Member, Officer) for any user	Admin only
Invite new members	Admin generates invite links or sends email invitations to new members with temporary credentials	Admin only
Deactivate accounts	Admin can deactivate a member's account without deleting their data; prevents login	Admin only
Member activity log	View last login, total logins, pitches voted on, and research viewed per member	Admin only
Member leaderboard	Optional gamified leaderboard showing individual stock pick performance for members who opt in	Calculated from pitch history

12. Educational Resources

A library of educational content to help members learn quantitative finance concepts and use the platform's research tools effectively.

Element	Description	Data Source / Notes
Model tutorials	Step-by-step guides on how to use each research model on the platform (DCF walkthrough, regression interpretation, reading comps)	Static content, officer-created
Financial statement guide	Interactive guide to reading income statements, balance sheets, and cash flow statements with real examples from holdings	Static content
Glossary	Searchable glossary of financial terms used throughout the platform	Static content
Resource library	Curated links to external books, courses, and articles recommended by officers	Officer/Admin managed
Video archive	Embedded or linked recordings of past club presentations and workshops	File storage or YouTube embed

13. Admin Panel

A dedicated admin control center accessible only to the admin account, providing full platform management capabilities.

Element	Description	Data Source / Notes
Dashboard overview	Admin-specific dashboard with platform health metrics: active users, content submissions this week, and system status	Internal analytics
Audit log	Complete log of all actions taken on the platform: logins, trades, content changes, and role changes with timestamps and actor	Internal database
Content moderation	Ability to edit or delete any content on the platform: research reports, pitch comments, meeting notes	Admin only
Platform settings	Configuration for portfolio benchmark selection, default model parameters, notification preferences, and theme settings	Internal config
Data export	Export portfolio data, trade history, and member activity as CSV or PDF for club records and reporting	Admin only
API key management	Manage API keys for market data providers and other external services	Admin only

14. Macro Dashboard

A dedicated section tracking macroeconomic indicators and market-wide data that provides context for the club's investment decisions.

Element	Description	Data Source / Notes
Interest rates	Current Fed Funds rate, 2Y/10Y Treasury yields, and yield curve chart with inversion indicator	Economic data API
Inflation data	Latest CPI, Core CPI, and PCE readings with historical trend chart	Economic data API (FRED)
Employment	Latest unemployment rate, non-farm payrolls, and initial jobless claims	Economic data API (FRED)
Market indices	Real-time display of S&P 500, NASDAQ, Dow Jones, Russell 2000, and VIX with daily change	Market data API
Economic calendar	Upcoming economic events: FOMC meetings, CPI releases, jobs reports, GDP releases, with dates and consensus estimates	Economic calendar API
Fear & Greed index	Visual gauge or indicator of current market sentiment	CNN Fear & Greed or custom
Sector performance	Heatmap or bar chart of sector ETF performance over 1D, 1W, 1M, and YTD	Market data API

15. Data Sources & API Layer

ICQ Labs will integrate with multiple data providers and expose an internal API for flexibility and extensibility.

15.1 External Data Sources

Element	Description	Data Source / Notes
Market data	Real-time and historical stock prices, volume, and market cap	Yahoo Finance API, Alpha Vantage, or Polygon.io
Financial statements	Income statement, balance sheet, and cash flow data for fundamental analysis	Financial Modeling Prep, Finnhub, or SEC EDGAR
News	Real-time news articles filtered by ticker	Finnhub, NewsAPI, or Benzinga
Economic data	Macroeconomic indicators (rates, CPI, employment)	FRED API

Options data	Options chain, IV, and Greeks (if options module is implemented)	CBOE, Tradier, or Polygon.io
SEC filings	Real-time alerts for new SEC filings	SEC EDGAR full-text search API

15.2 Internal API

Element	Description	Data Source / Notes
REST API	Internal API endpoints for all platform data (holdings, research, pitches, members) with role-based authentication	Backend framework
Notebook access	Documented API so members can pull club portfolio data into their own Python/Jupyter notebooks for custom analysis	API documentation page
Webhooks	Optional webhook integrations for Slack/Discord notifications on new pitches, trades, or earnings alerts	Webhook service
PDF report export	API endpoint to generate formatted PDF reports of portfolio performance, individual stock research, or pitch summaries	PDF generation service